Do ER status and positive nodes relate to 5-year death in the same way in SEER and METABRIC?

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STA510 Bayesian statistical methods

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1 Introduction

SEER is a US cancer registry [1, 2] and METABRIC is a UK/Canada cohort [3, 4]; the two come from different countries and periods. ER status and the number of positive lymph nodes are established prognostic factors [5, 6], so it is natural to ask whether their associations with survival transfer between these populations. An earlier, unadjusted analysis of the same datasets [7] found significant differences between them, but it used different outcomes (survival months and final status) and unequal follow-up (up to 107 months in SEER and 355 months in METABRIC), which leaves open whether the associations really differ.

This report extends that analysis in three ways: it uses one common outcome, death within five years; it adjusts for age, tumour size and grade; and it reports the size of the difference with its uncertainty, not only whether it is significant.

1.1 Research question

Are the associations of ER status and positive nodes with death within five years the same in SEER and METABRIC, and by how much do they differ? Let β_{SEER} and β_{METABRIC} be the log-odds coefficients of ER or of nodes in each dataset. The difference is defined as

$$\delta = \beta_{\text{METABRIC}} - \beta_{\text{SEER}},$$

one for ER and one for nodes. Then $\exp(\delta)$ is the ratio of the odds ratios in the two datasets: $\exp(\delta) = 1$ means the same association, and $\exp(\delta) = 2$ means one odds ratio is twice the other. The answer is the posterior distribution of δ.

2 Data

2.1 Variables

Table 1 lists the variables. Age, tumour size and tumour grade are adjustment variables: they are in the model in both datasets, but the research question concerns only ER and nodes. In SEER the grade label “anaplastic” is coded as grade 3.

Variable	Description
ER status	Tumour cells have estrogen receptors (yes/no)
Positive nodes	Lymph nodes with cancer cells (count)
Age	At diagnosis (years)
Tumour size	Size at diagnosis (mm)
Tumour grade	How abnormal the cells look (1–3)
Survival months	Follow-up time from diagnosis
Status	Alive or dead at last follow-up

Table 1: Variables used. Survival months and status define the outcome.

2.2 Outcome and patient selection

The outcome is death within five years (60 months): $y_i = 1$ if patient i died within 60 months of diagnosis and $y_i = 0$ otherwise. Final status is not used because METABRIC follows patients far longer than SEER, so late deaths would make the datasets incomparable. Table 2 shows how many patients were removed and why.

	SEER	METABRIC
Raw patients	4,024	2,509
No status or time (outcome cannot be built)	−19	−528
Alive with < 60 months of follow-up (outcome unknown)	−753	−64
Missing covariates (model needs all five)	0	−157
Restriction to age 30–69 and ≥ 1 positive node	–	−1,184
Analysed	3,252	576

Table 2: Patient selection. Deaths are always kept; the restriction in METABRIC makes it resemble SEER.

Patients alive with fewer than 60 months of follow-up have an unknown five-year outcome and are dropped (complete cases). This rule is an assumption: deaths are always observed while some early-censored patients would have died, so absolute death rates are probably somewhat too high. The share dropped for this reason differs between the datasets (753 of 4,005 in SEER and 64 of 1,917 in METABRIC), so the distortion may not cancel in the comparison; it would matter if it differed between ER groups. Survival models would handle censoring properly but are outside the methods of the course.

2.3 Descriptive summary

Crude five-year death rates (not adjusted) are higher for ER-negative patients and rise with the number of positive nodes in both datasets (Table 3). The ER gap is about the same in percentage points in the two datasets (26.8 in SEER, 27.5 in METABRIC) and smaller in relative terms in METABRIC. The METABRIC 7-or-more band probably rests on few patients (100 in the analysed sample), so it is noisy. Table 8 in Appendix A gives the characteristics before restriction.

Crude 5-year death rate	SEER	METABRIC
ER−	38.4%	47.1%
ER+	11.6%	19.6%
1 positive node	8.5%	17.3%
2–3 nodes	9.1%	27.7%
4–6 nodes	16.3%	33.7%
7 or more nodes	28.0%	54.0%

Table 3: Crude five-year death rates in the analysed samples (3,252 SEER and 576 METABRIC patients).

3 Methods

3.1 Model

The outcome is modelled with Bayesian logistic regression:

$$y_i \sim \text{Bernoulli}(\mu_i),$$
$$\text{logit}(\mu_i) = \alpha_{j(i)} + \beta_{1,j(i)} X_{1i} + \beta_{2,j(i)} X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i},$$

where $j(i)$ is the dataset of patient i, X_1 is ER status (ER+ versus ER−), $X_2 = \log(1 + \text{nodes})$, X_3 is age, X_4 is tumour size and X_5 is grade. The intercept α, the ER effect β_1 and the nodes effect β_2 are separate per dataset, because that is what the question compares. Age, size and grade (β_3, β_4, β_5) are shared; this is an assumption that model M2 tests below. Covariates are centred at age 54 years, tumour size 25 mm, grade 2 and 2 positive nodes (on the $\log(1 + \text{nodes})$ scale), so the intercept refers to a reference patient. There are nine parameters in total. The dataset difference is $\delta = \beta_{j=\text{METABRIC}} - \beta_{j=\text{SEER}}$, computed for ER ($\beta_1$) and for nodes ($\beta_2$).

3.2 Priors

Weakly informative normal priors were built from published studies (Table 4). For each prior, a published rate or ratio was turned into an odds ratio, the log per unit of the covariate was taken, the value was shrunk because the source is unadjusted or refers to another time horizon, and the resulting range was turned into a centre and a standard deviation. The shrinkage step is a judgement. The sources are not from identical conditions to the data used here. The same priors are used for SEER and METABRIC.

Parameter	Prior $N(\text{mean}, \text{sd})$	Source
α, intercept	$N(-1.815,\ 1)$	NCI survival statistics [8]
β_1, ER+ vs ER−	$N(-0.57,\ 0.5)$	Bentzon et al. [5]; Moffat [6]
β_2, $\log(1 + \text{nodes})$	$N(0.7,\ 0.4)$	Moffat [6], fig. 28-3 (SEER data)
β_3, age (per year)	$N(0.0203,\ 0.0101)$	Kim et al. [9]; Dai et al. [10]
β_4, size (per mm)	$N(0.0265,\ 0.0133)$	Moffat [6], fig. 28-3 (SEER data)
β_5, grade (per step)	$N(0.4,\ 0.25)$	El Saghir et al. [11]

Table 4: Priors and their sources.

The prior on α corresponds to a baseline five-year death rate of about 14% ($\text{logit}^{-1}(-1.815)$). Before fitting, the death rates implied by the priors alone were simulated (a prior predictive check, Section 4.3).

3.3 Models compared

The question is how much the two datasets should share. Four models are compared (Table 5).

Model	ER and nodes	Age, size, grade	Pooling
M0	one shared coefficient ($\delta = 0$)	shared	complete
MH	$\beta_j = \beta_0 + u_j$, $u_j \sim N(0, \sigma^2)$, $\sigma \sim$ Exponential(2)	shared	partial
M1 (main)	separate per dataset	shared	none
M2	separate per dataset	separate per dataset	none

Table 5: Models compared. α is separate per dataset in all models.

M0 assumes ER and nodes work identically in the two datasets. M1 lets them differ freely. MH lies between: the datasets share a common mean β_0 and deviate from it by u_j, with a spread σ that the data estimate; the deviations are independent for ER and nodes. When σ is near zero MH approaches M0, and when σ is large it approaches M1. The prior $\sigma \sim$ Exponential(2) has mean 0.5 and median about 0.35. Because $\delta = u_{\text{METABRIC}} - u_{\text{SEER}}$ is the difference of two independent deviations with standard deviation σ, its standard deviation is $\sigma\sqrt{2}$, about $0.5\sqrt{2} \approx 0.71$ for a typical σ. In M1, the independent priors $N(\cdot, 0.5)$ on the two ER coefficients give a prior sd of $0.5\sqrt{2} \approx 0.71$ for δ as well. The two models therefore start with a similar view of how different the datasets could be. M2 asks a different question: whether age, size and grade also need dataset-specific slopes.

For each posterior draw, δ is the difference between the METABRIC and SEER coefficients (M1 and M2), or between the two group-level deviations (MH, where β_0 cancels). The draws of δ are summarised by their median and 95% interval. M0 has no δ because it is zero by assumption.

3.4 Fitting

The posterior has no closed form, so it is sampled with Hamiltonian Monte Carlo (the No-U-Turn sampler in Stan, through the `brms` package [12]). Each model was fitted with the default `brms` settings: 4 chains of 2,000 iterations, half of them warm-up, giving 4,000 draws. The exception is `adapt_delta`, which was set to 0.95 and raised to 0.99 for MH after 9 divergent transitions. Convergence was judged with $\hat{R}$ (rule: below 1.1), effective sample size (ESS), divergences and trace plots.

3.5 Model checking and comparison

Convergence says that the sampler works for a given model, not that the model fits the data. The model is therefore checked in three ways.

- **Prior predictive check:** the death rates implied by the priors alone, before seeing the data.
- **Posterior predictive checks:** for each posterior draw a 0/1 outcome is simulated for every patient and the death rate is computed in a group of patients (for example, SEER ER+ patients or METABRIC patients aged 55–64). The distribution of these replicated rates is compared with the observed rate. The tail probability p_{upper} is the share of replications at or above the observed rate; following the course notes [13], values below about 0.11 or above 0.89 suggest that the model needs extending.
- **Leave-one-out cross-validation (loo):** the expected log predictive density (elpd) estimates how well each model predicts a patient not used in the fit. It is computed by Pareto-smoothed importance sampling from one fit [14], and differences in elpd are read against their standard errors.

4 Results

4.1 Dataset differences

Figure 1 and Table 6 show the posterior of $\delta = \beta_{\text{METABRIC}} - \beta_{\text{SEER}}$. All six 95% intervals include 0, so none of the models shows a clear difference between the datasets.

ER status. Under M1 the median δ is 0.15, so $\exp(\delta) = 1.16$: the odds ratio of ER+ versus ER− is estimated to be 16% higher in METABRIC than in SEER. Since this odds ratio is below 1, a higher value means that ER+ is somewhat less protective in METABRIC. The interval is wide (0.73 to 1.86), however. The data are compatible with an odds ratio that is 27% lower in METABRIC, and also with one that is almost twice as high, so a moderate difference cannot be excluded.

Positive nodes. The median δ is 0.07 ($\exp(\delta) = 1.08$, 95% interval 0.81 to 1.43), which leads to the same conclusion with less uncertainty.

Other models. Letting age, size and grade differ by dataset (M2) does not change these results, and neither does partial pooling (MH). The MH intervals are narrower because MH assumes that the datasets are similar and pulls their coefficients towards each other. The data say little about how similar they are: the posterior median of σ is 0.18 for ER and 0.13 for nodes, but its 95% upper limits are 1.12 and 0.78, since only two datasets are available to estimate it. The narrower MH intervals therefore reflect the model's assumption more than additional information.

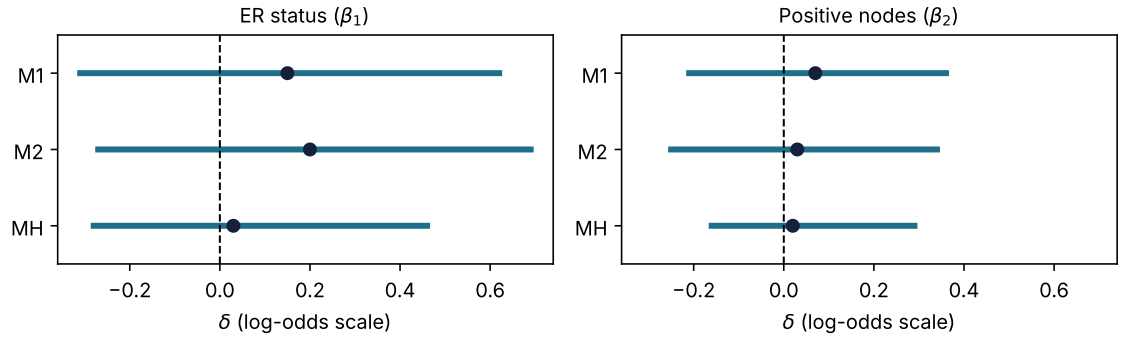


Figure 1: Posterior median (dot) and 95% interval (bar) of δ for ER status and positive nodes under M1, M2 and MH. The dashed line is $\delta = 0$.

Effect	Model	Median δ	95% interval	$\exp(\delta)$ (95%)	$P(\delta > 0)$
ER	M1	0.15	−0.31 to 0.62	1.16 (0.73 to 1.86)	0.74
ER	M2	0.20	−0.27 to 0.69	1.23 (0.76 to 1.99)	0.80
ER	MH	0.03	−0.28 to 0.46	1.03 (0.76 to 1.59)	0.62
Nodes	M1	0.07	−0.21 to 0.36	1.08 (0.81 to 1.43)	0.70
Nodes	M2	0.03	−0.25 to 0.34	1.03 (0.78 to 1.40)	0.59
Nodes	MH	0.02	−0.16 to 0.29	1.02 (0.85 to 1.34)	0.64

Table 6: Posterior summaries of δ (METABRIC − SEER) on the log-odds scale.

4.2 Convergence

All models converged. $\hat{R}$ was 1.00; the effective sample size exceeded 2,800 for M1 and M2, and the lowest value for MH was about 1,050. After raising `adapt_delta` to 0.99 for MH there were no divergences, and trace plots showed well-mixed chains (Appendix C).

4.3 Prior predictive check

Simulated from the priors alone over all patients (4,000 draws), the implied five-year death rate has a median of 24.0% in SEER (95% range 4.9–62.6%) and 27.6% in METABRIC (95% range 6.2–66.3%). This range is wider than the 10–25% observed in real cohorts, but it contains the plausible rates and excludes the absurd ones: death rates close to 0% or 100% receive almost no prior probability. The priors are therefore weakly informative, and with 3,828 patients the data dominate the posterior.

4.4 Posterior predictive checks

The group-rate check was run on M1 for all five covariates in both datasets. For ER, nodes and grade, all observed rates lie inside the 95% range of the replications (Appendix B). The most extreme group is SEER with 2–3 nodes ($p_{\text{upper}} = 0.92$, just beyond the 0.89 rule of thumb): the observed rate rises little between 1 node and 2–3 nodes, which is flatter than the model predicts. This is a mild shape issue. Tumour size and age show clearer problems. For **tumour size** (Figure 2), observed death rates are below the replications for tumours of 15 mm or less and above them for 26–40 mm, in both datasets. For **age** (Figure 3), young METABRIC patients (30–44) die more often than predicted ($p_{\text{upper}} = 0.021$), METABRIC patients aged 55–64 less often (0.972), and SEER patients aged 65–69 more often (0.046); letting the age slope differ by dataset (M2) does not remove this. The pattern is in the same direction in both datasets, which suggests that the linear terms for size and age are approximations and that the true relationships are non-linear (a U-shape for age). Because the problem is not specific to one dataset, it is unlikely to explain a difference between them, although this has not been tested.

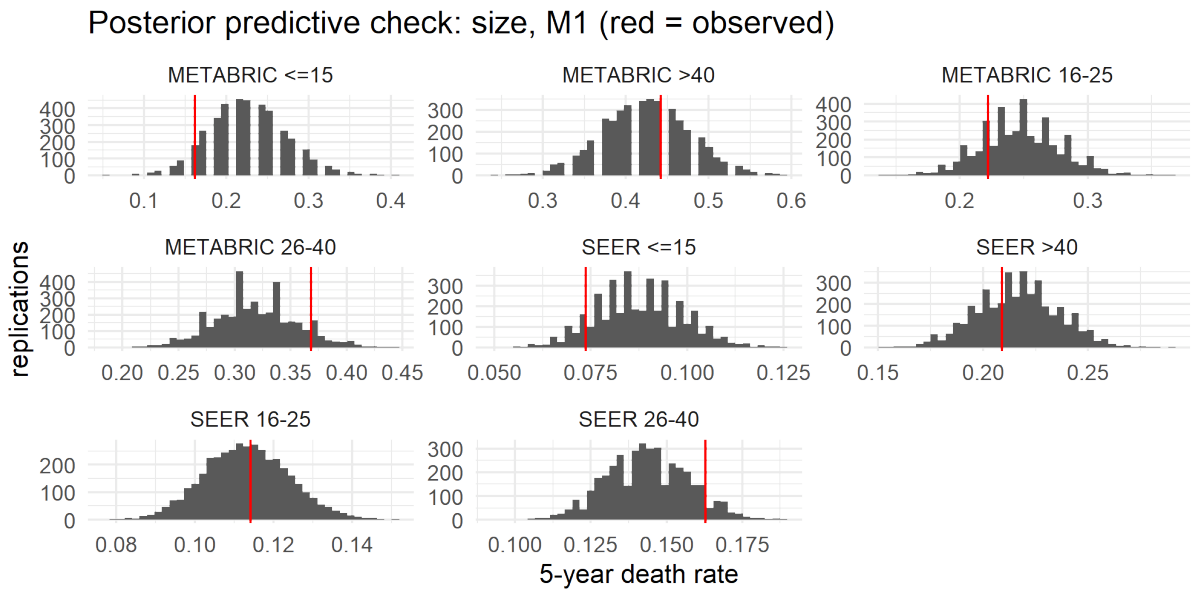


Figure 2: Posterior predictive check by tumour size band under M1. Bars: replicated death rates; red line: observed rate.

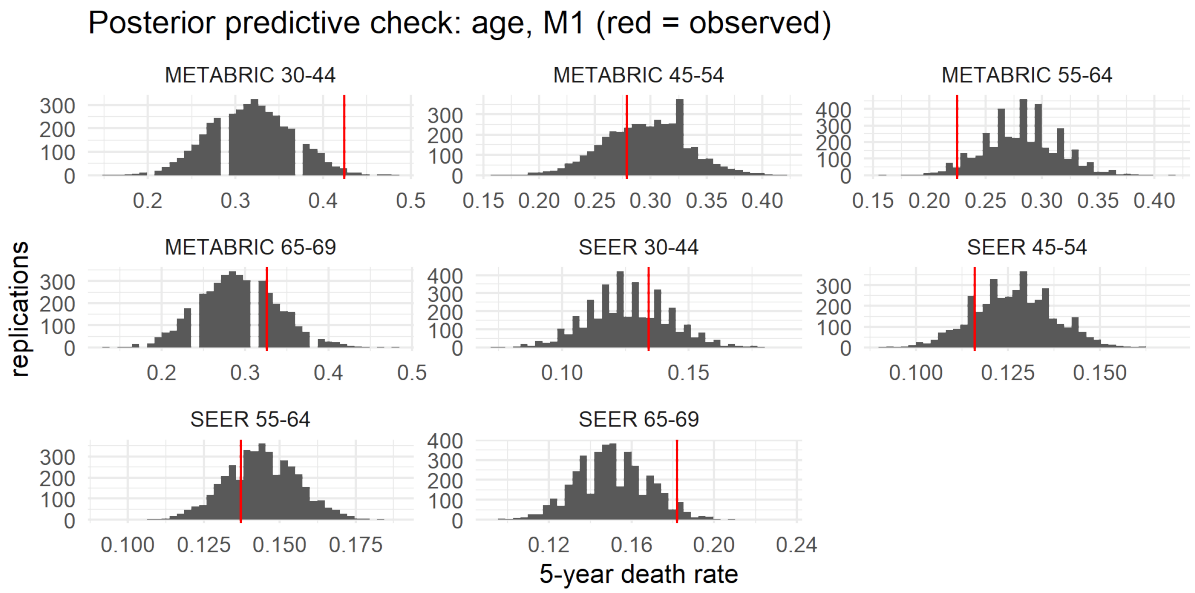


Figure 3: Posterior predictive check by age band under M1.

The check was also run for ER and nodes under M0, which forces the same effects in both datasets. M0 reproduces the observed rates about as well as M1 and no group moves into a tail (Appendix B), which is consistent with there being no clear dataset difference; with 576 METABRIC patients, however, the check has little power to detect a moderate difference.

4.5 Model comparison

Table 7 gives the loo comparison. All differences in elpd are below 4, so the models predict about equally well; the Pareto k values are below 0.7 for all four models, so the approximation is reliable. The largest difference, M1 versus M0 (-1.7, se 0.8), is about two standard errors but small in absolute terms, so it is weak support for pooling and not a clear win. Separate slopes for age, size and grade give no predictive gain (M2 versus M1: -0.2, se 1.8), so M1 is kept as the main model. For M1, $\text{elpd}_{\text{loo}} = -1458.4$ (se 37.6) with an effective number of parameters $p_{\text{loo}} = 8.2$.

Pooling of ER and nodes	Model	elpd_diff (se)
Complete	M0	0
Partial	MH	-0.9 (0.4)
None	M1	-1.7 (0.8)
Age, size, grade	**Model**	**elpd_diff (se)**
Shared	M1	0
Separate	M2	-0.2 (1.8)

Table 7: Leave-one-out comparison. Higher elpd is better; elpd_diff is the difference to the reference model of each comparison.

5 Discussion

5.1 Interpretation

There is no clear evidence that the association of ER status or of positive nodes with five-year death differs between SEER and METABRIC: all intervals for δ include zero, in all three models, and the model checks and loo are consistent with this. This is not evidence that the associations are the same. For ER a ratio of odds ratios of up to about 2 is compatible with the data, and the posterior is only mildly lopsided ($P(\delta > 0) = 0.74$ under M1). The conclusion does not depend on the modelling choice: complete pooling, partial pooling, no pooling and separate adjustment slopes give the same answer. In MH the narrower intervals partly reflect the assumption that the datasets are similar. The significant differences found in the earlier study may come from the unequal follow-up and the lack of adjustment, which is plausible but was not tested here.

5.2 Limitations

- **Censoring.** Excluding patients with an unknown five-year outcome assumes that censoring is unrelated to the outcome, and the excluded share differs between datasets. Absolute death rates are probably somewhat too high.
- **Functional form.** The linear age and size terms fit only approximately (and the nodes term possibly mildly); the posterior predictive checks show a non-linear pattern.
- **Two groups only.** With two datasets, the data say little about how much to pool (σ), so the MH results depend on its prior.
- **Priors.** They are anchored to literature that is often unadjusted and from different populations. A prior sensitivity analysis was not run.
- **Restriction of METABRIC.** Restricting to age 30–69 and at least one node makes METABRIC resemble SEER but discards most of its patients, and the analysis has not been repeated on the unrestricted sample. METABRIC has few patients in some groups.

5.3 Next steps

Model log tumour size and a flexible age term; run a prior sensitivity analysis (for example other rates of the exponential prior on σ and narrower or wider priors on the coefficients); repeat the analysis with the unrestricted METABRIC sample.

6 Conclusion

In a Bayesian logistic regression of death within five years in 3,252 SEER and 576 METABRIC patients, adjusted for age, tumour size and grade, there is no clear evidence that the associations of ER status and positive nodes with death differ between the datasets. A moderate difference is not excluded: for ER, a ratio of odds ratios of up to about 2 is compatible with the data (95% interval 0.73 to 1.86 under M1), so the result should not be read as showing that the associations are the same.

A Characteristics before restriction

Before restriction	SEER	METABRIC
Patients	3,252	1,760
Died within 5 years	442 (13.6%)	398 (22.6%)
ER-positive share	92.6%	76.3%
Death rate, ER+	11.6%	17.6%
Death rate, ER−	38.4%	38.8%
Age, median (quartiles)	54 (47–62)	61.7 (51.3–70.6)
Positive nodes, median (quartiles)	2 (1–5)	0 (0–2)
Tumour size mm, median (quartiles)	25 (16–39)	23 (17–30)

Table 8: Patient characteristics before the METABRIC restriction (SEER is unchanged). The analysed METABRIC sample has 576 patients, with death rates of 47.1% (ER−) and 19.6% (ER+).

B Posterior predictive checks for ER, nodes and grade

Figures 4–6 show the M1 checks and Figure 7 the M0 checks. Tables 9–11 give the observed rates and p_{upper}. The METABRIC grade 1 group has only 36 patients, so its replicated rate takes few values.

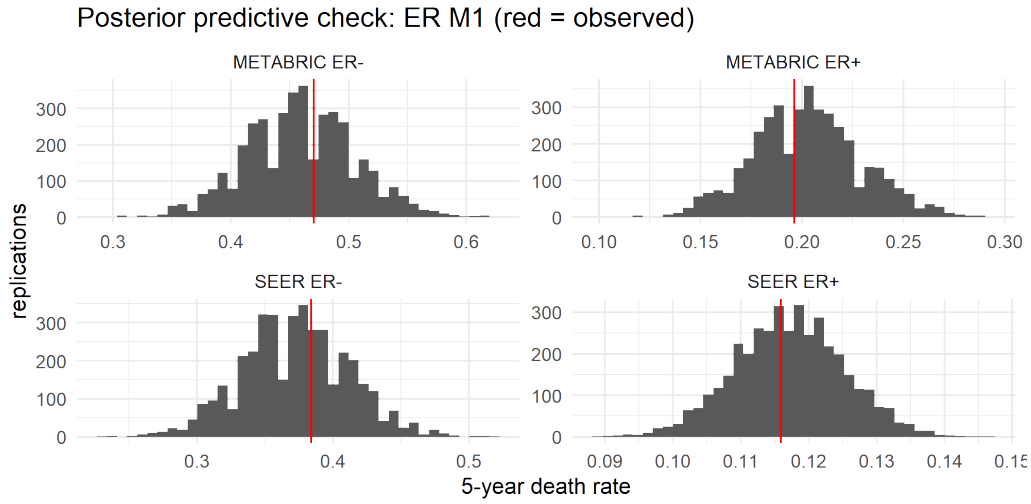


Figure 4: Posterior predictive check by ER status under M1.

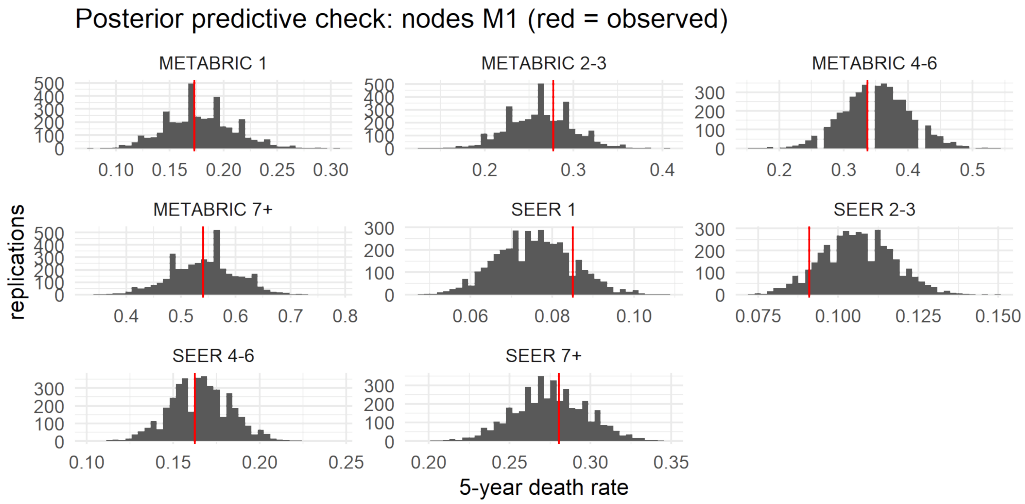


Figure 5: Posterior predictive check by number of positive nodes under M1.

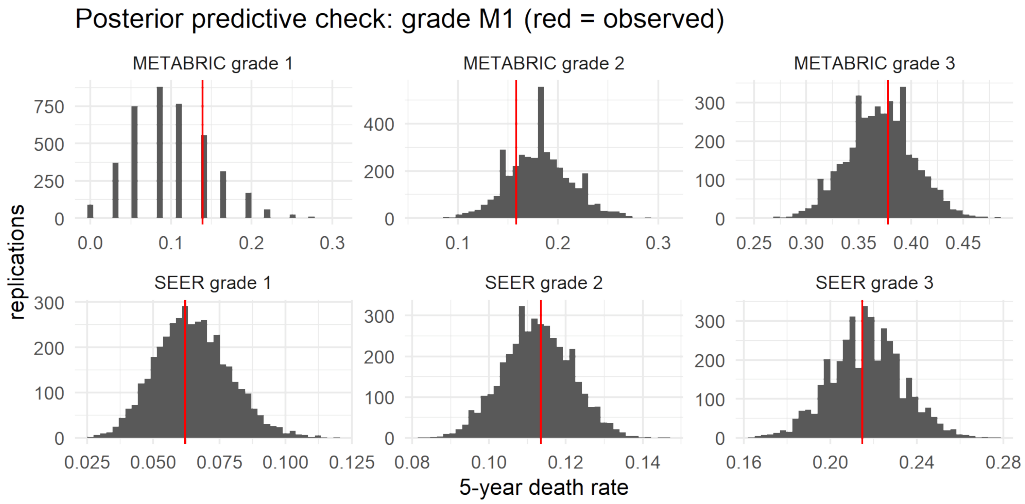


Figure 6: Posterior predictive check by tumour grade under M1.

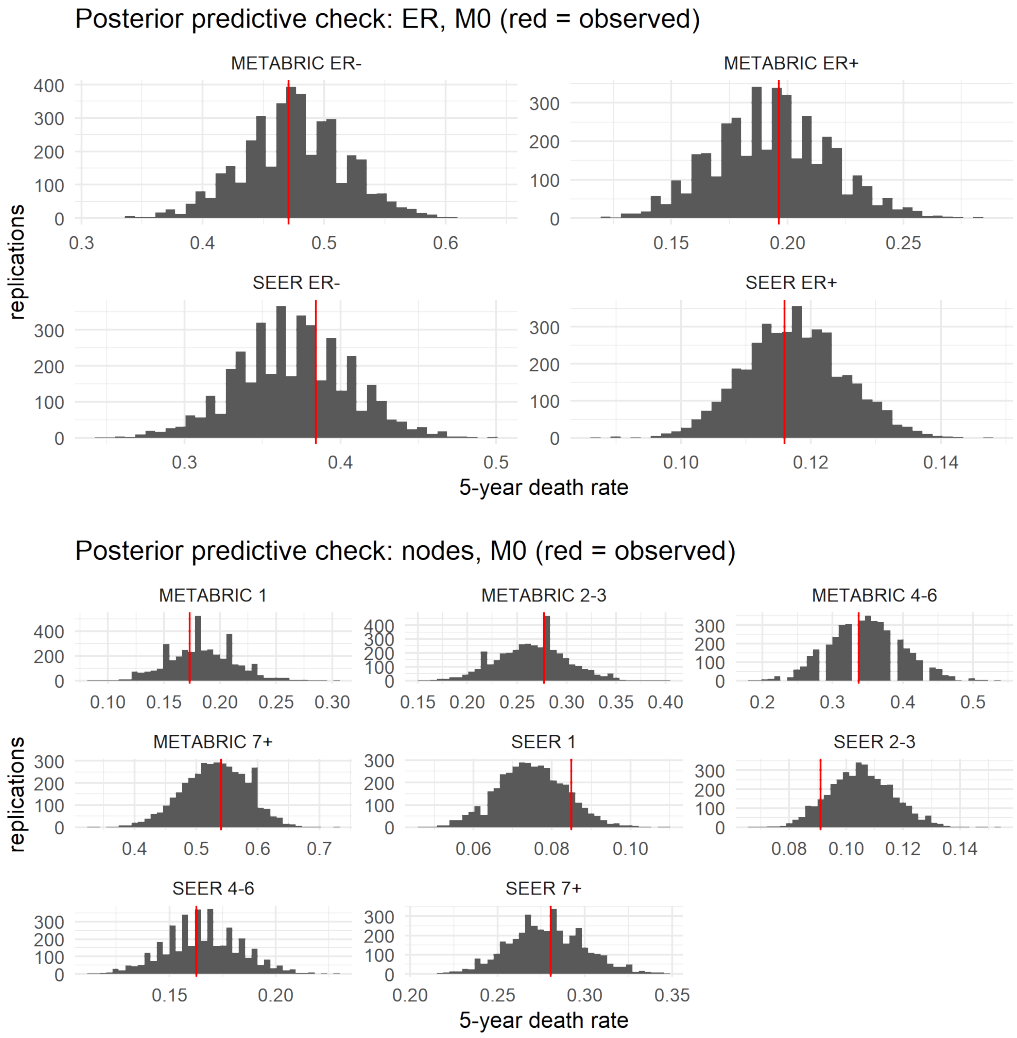


Figure 7: Posterior predictive checks for ER (top) and nodes (bottom) under M0.

Group	n	Observed	Replicated median (95%)	p_{upper} M1	p_{upper} M0
METABRIC ER−	204	0.471	0.461 (0.368–0.554)	0.423	0.577
METABRIC ER+	372	0.196	0.202 (0.151–0.258)	0.601	0.471
SEER ER−	242	0.384	0.372 (0.298–0.450)	0.399	0.363
SEER ER+	3010	0.116	0.117 (0.102–0.133)	0.569	0.573

Table 9: ER groups: observed death rate, M1 replications, and p_{upper} under M1 and M0.

Group	n	Observed	Replicated median (95%)	p_{upper} M1	p_{upper} M0
METABRIC 1	214	0.173	0.178 (0.117–0.243)	0.596	0.668
METABRIC 2–3	173	0.278	0.260 (0.191–0.335)	0.364	0.400
METABRIC 4–6	89	0.337	0.348 (0.247–0.449)	0.647	0.616
METABRIC 7+	100	0.540	0.540 (0.420–0.660)	0.558	0.458
SEER 1	1212	0.085	0.075 (0.059–0.094)	0.172	0.141
SEER 2–3	924	0.091	0.106 (0.084–0.129)	0.916	0.910
SEER 4–6	492	0.163	0.165 (0.132–0.199)	0.582	0.596
SEER 7+	624	0.280	0.276 (0.231–0.321)	0.419	0.463

Table 10: Node bands: observed death rate, M1 replications, and p_{upper} under M1 and M0.

Group	n	Observed	Replicated median (95%)	p_{upper} M1
METABRIC grade 1	36	0.139	0.083 (0.028–0.194)	0.285
METABRIC grade 2	183	0.159	0.180 (0.120–0.246)	0.797
METABRIC grade 3	357	0.378	0.370 (0.308–0.434)	0.431
SEER grade 1	419	0.062	0.064 (0.041–0.093)	0.598
SEER grade 2	1902	0.114	0.113 (0.095–0.130)	0.463
SEER grade 3	931	0.215	0.217 (0.184–0.250)	0.564

Table 11: Grade groups: observed death rate, M1 replications, and p_{upper} under M1.

C Trace plots

Figures 8–11 show the posterior densities and trace plots of all parameters for M1 (main model) and MH (the model that needed `adapt_delta` 0.99). The four chains overlap and show no trends or stuck periods. M2 and M0 gave the same picture and are not shown.

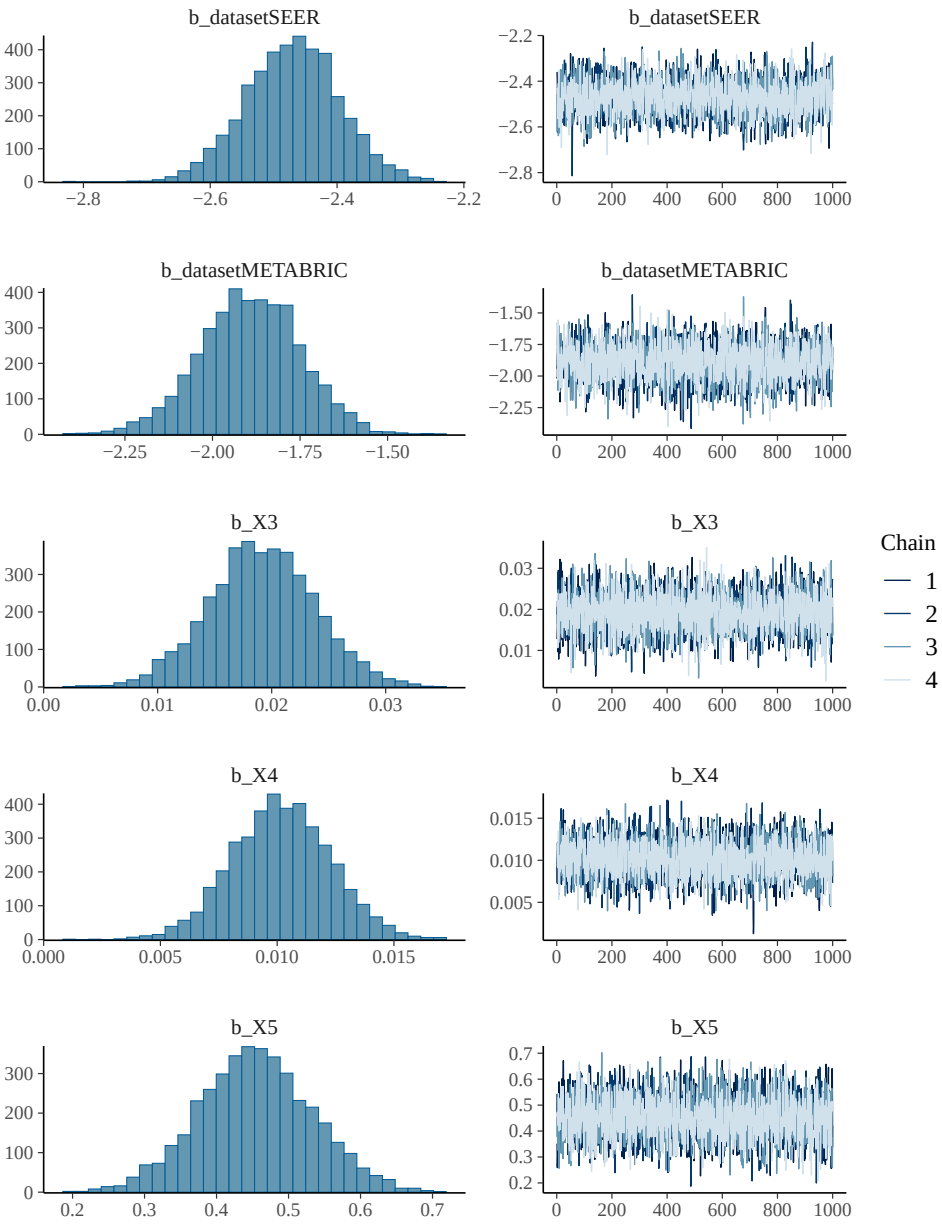


Figure 8: M1: densities and trace plots (part 1 of 2).

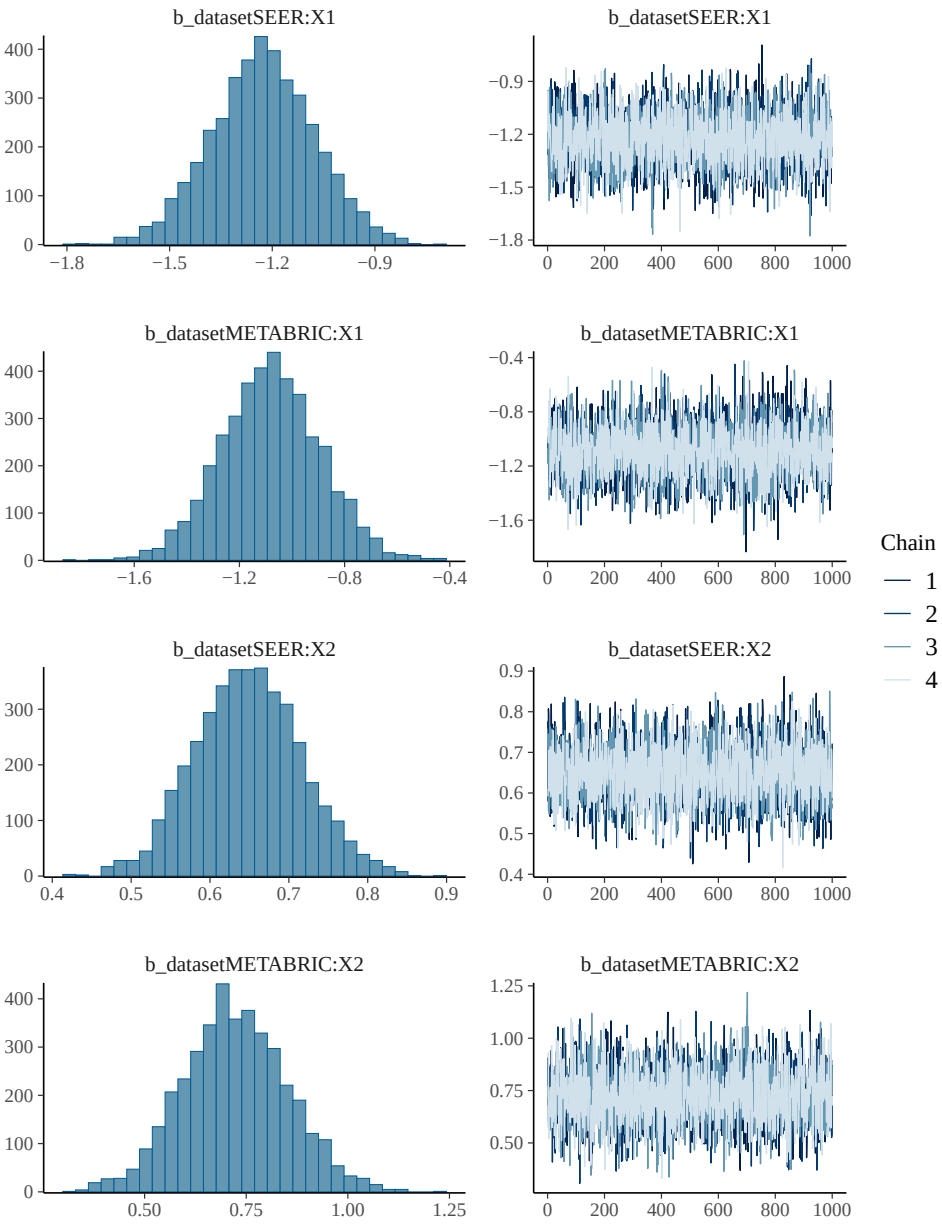


Figure 9: M1: densities and trace plots (part 2 of 2).

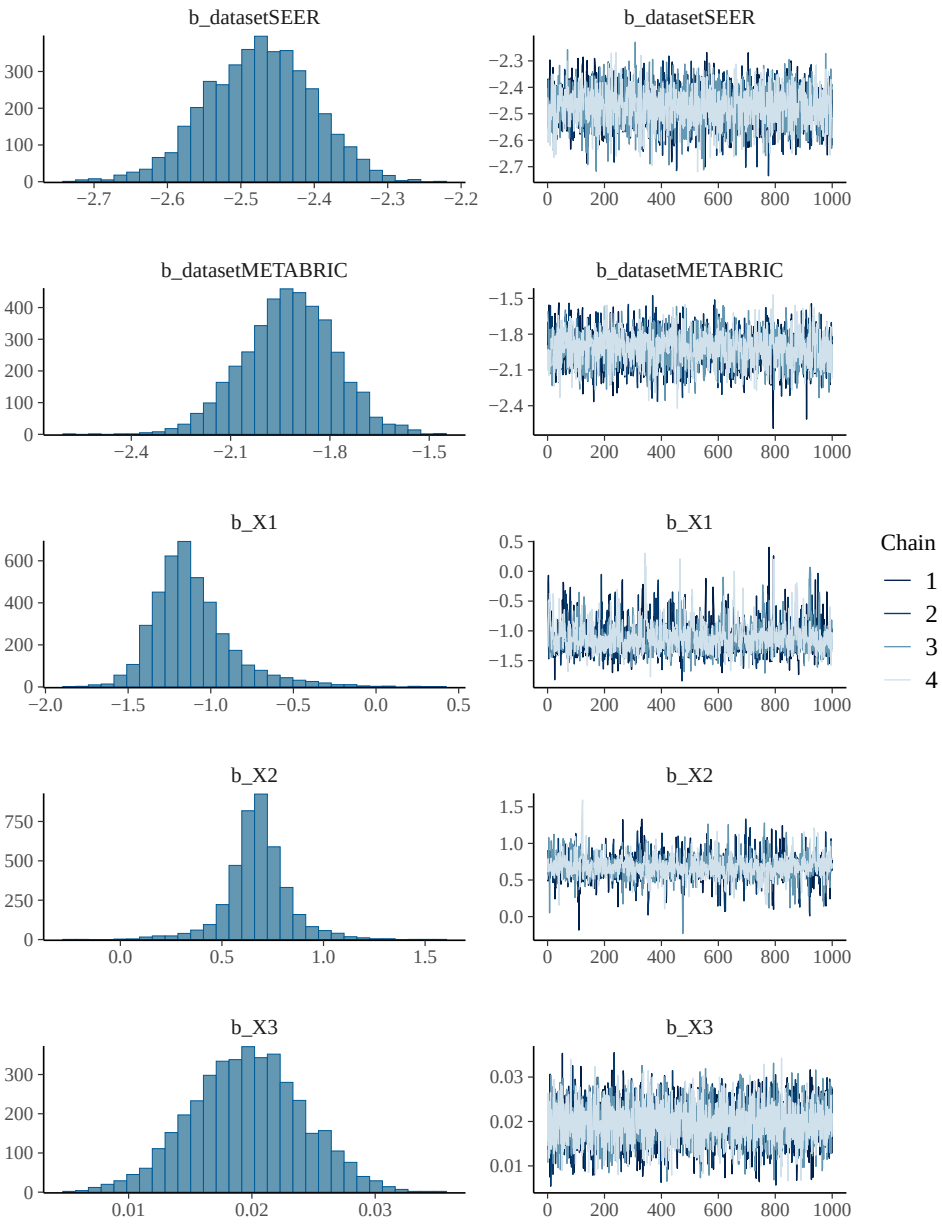


Figure 10: MH: densities and trace plots (part 1 of 2).

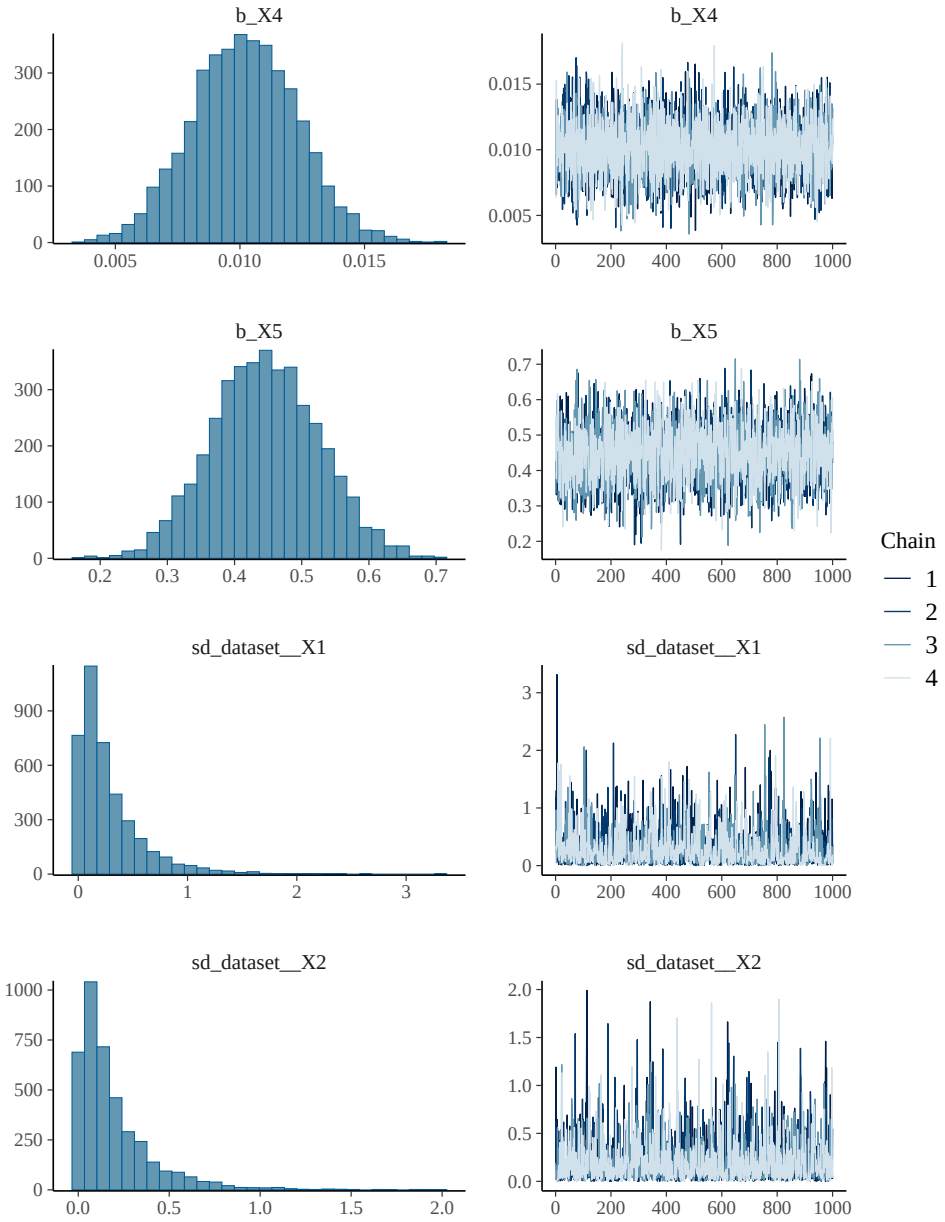


Figure 11: MH: densities and trace plots (part 2 of 2).

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